

Harsath Kaliappa Gounder Thangavel

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Education

York University

Bachelor of Science in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

Work Experience

Privex

Software Engineer

October 2023 – January 2024

London, United Kingdom

- Developed an AI customer service agent using RAG techniques on a self-hosted LLM to handle common customer inquiries related to technical issues, products, and services.
- Developed REST APIs in Flask for querying and training the LLM, integrated with Discord APIs to allow customers to query the AI agent and staff to train it with new data on demand.
- Utilized vector databases and MySQL for training data management. Developed scripts to remove existing data from the database and to add new data from the website's documentation.
- Developed systemd service files and scripts to monitor and automatically reboot for error recovery.

CogniPet

Software Engineer

April 2019 – December 2019

Zürich, Switzerland

- Developed Pet Breed Detection system by training a deep learning model using the Xception neural network and transfer learning techniques in Keras to recognize 120 breeds of dogs.
- Implemented REST APIs in Flask for model inference and integrated them with CogniPet's Android and iOS apps using Amazon Web Services (AWS).
- Experimented with video-based techniques for a pet facial recognition system by collecting data from dogs and analyzing features from multiple frames.

Projects

Jix Interpreter | C Interpreter

- Developed an interpreter for a dynamically typed programming language.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented basic functional programming support, including first-class and higher-order functions.
- Tested using automated tests with CI/CD pipelines.

Prism Coverage | Java Code Coverage Engine

- Developed a code coverage engine using ANTLR4 for a programming language.
- Implemented control flow coverage (statement, decision, condition, path) criteria.
- Implemented data-flow coverage (All-Defs, All-DU-Paths, All-P-Uses) criteria.
- Implemented color-coded HTML report generation for the coverage engine.

ToDo App | Flutter Mobile App

- Developed a cross-platform ToDo application using Flutter and Dart for both Android and iOS.
- Implemented task management features with priority-based sorting and local storage using Sqflite.
- Created responsive UI/UX for task list, task addition, and exit confirmation screens.
- Built and tested the app for both Android and iOS environments, with APK deployment for Android.

Blueth | C++ Networking Library

- Developed a highly concurrent and event-driven networking framework for Linux.
- Implemented event-loop abstractions, HTTP parsers, proxies, thread-pool executors, SIMD and codec algorithms.
- Implemented thread-safe cache containers and synchronization mechanisms for multithreaded environments.
- Tested using automated tests with CI/CD pipelines.

Technical Skills

Languages: Python, C++, C, Java, SQL, JavaScript, Go, Perl, Bash, PHP, HTML/CSS, $\text{\LaTeX}$
Technologies: Linux/Unix, CI/CD, MySQL, PostgreSQL, Docker, Kubernetes, Redis, MongoDB, JWT, OpenAPI
Developer Tools: Git, Linux Containers, GDB, Amazon Web Services
Frameworks: Node.js, Flutter, Flask, Django, JUnit